PROJECT DOCUMENTATION

Bank Transaction Fraud Detection: Rule-Based Fraud Analytics

Industry-Oriented Data Analytics Student Project

Rule-Based Bank Transaction Fraud Detection & Suspicious Activity Analytics Using Python, Pandas, NumPy, Matplotlib and FastAPI

Item	Details
Project group	Fraud Analytics Group / Team Cronus
Team members	Student Data Analyst (Portfolio Capstone)
Course / Programme	Industry-Oriented Data Analytics Capstone / Financial Analytics
Faculty guide	Fraud Risk Operations & Applied AI Guide
Technology	Python 3.11, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn (analysis); FastAPI, Tailwind CSS, Chart.js, HTML5 (interactive dashboard)
Dataset	Bank transaction screening records, 10,000 raw rows x 13 columns (synthetic, calendar month September 2026)
Date of submission	November 2026
Disclaimer	The transaction dataset used in this project is synthetic and was created for learning and demonstration adhering to Section 7 privacy rules. All results describe this dataset and demonstrate an analytical workflow; they must not be read as evidence about a real banking operation or real customer accounts.

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Abstract

Financial institutions process millions of electronic banking transactions daily. Punctual, non-disruptive clearing of legitimate payments is essential to consumer trust, yet fraudulent card testing, account takeovers, and unauthorized wire transfers inflict catastrophic losses. This project analyses 10,000 raw transaction records from a synthetic banking network to implement an explainable, transparent, rule-based screening engine. Rather than automatically declaring customers as criminals, the system evaluates behavioral anomalies across Transaction Amount, Transaction Time, and Transaction Velocity, assigning an interpretable Suspicion Score (0 to 3) to triage records for human-in-the-loop compliance review.

The screening engine was implemented as a modular Python pipeline (loading, validation, metric derivation, rule evaluation, KPI calculation, multidimensional drilldowns, and charting) and extended with a responsive web dashboard (Cronus FraudGuard) and REST API. Across 10,000 evaluated transactions, 118 records (1.18%) triggered multiple co-occurring anomalies (Suspicion Score >= 2) and were escalated to the priority review queue. Flagged transactions averaged $53,416.15—a 13.47x surge over normal volume ($3,966.30). Transfer was identified as the highest-risk transaction type (3.17% suspicious rate) and Online as the highest-risk channel (2.56%). Against verified historical ground-truth fraud labels, the rule-based screener achieved 100.0% precision, 0.00% false positive rate, and captured all acute multi-factor attack patterns.

Keywords: fraud analytics, rule-based screening, suspicion score, transaction anomaly detection, Pandas, Matplotlib, FastAPI, banking compliance.

1. Project Overview

1.1 Problem Statement

Retail banks face escalating digital fraud threats across online banking, mobile applications, and payment gateways. While machine learning is frequently proposed, deploying black-box AI models as an unexplainable first-tier filter creates severe legal, regulatory (RBI / OCC / GDPR), and customer relations liabilities: compliance officers cannot explain to auditors or clients why an alert was triggered. A robust, deterministic, rule-based transaction screening system is required to screen transactions, provide clear audit trails, and prioritize manual investigation without prematurely freezing customer accounts.

1.2 Scope of the Work

- Ingestion, cleaning, and validation of 10,000 transaction records with zero data loss on valid keys.
- Computing statistical thresholds (Gaussian mean + 3 standard deviations and 99th percentile P_{99}).
- Evaluating three core behavioral heuristics: Amount Outlier, Late-Night Window (23:00 to 05:59), and Rolling 1-Hour Burst count.
- Combining rules into an explainable Suspicion Score (0 to 3) and 3-digit bitmasks (e.g. `1_1_0`).
- Producing 10 high-resolution Matplotlib figures and CSV investigation queues.

- Delivering an interactive web dashboard with dynamic threshold simulation and CSV upload capabilities.

1.3 Project Deliverables

All project brief deliverables are fulfilled:

1. Complete Python package (`src/`) and CLI runner (`src/cli.py`).
2. Cleaned primary dataset (`data/bank_transactions.csv`) and sample upload file (`test_upload_sample.csv`).
3. Priority manual review queue (`reports/suspicious_transactions.csv`).
4. Account risk summary profiles (`reports/account_risk_summary.csv`).
5. 10 publication-ready Matplotlib figures in `reports/figures/`.
6. Executive management report answering all 10 core business questions.
7. Interactive web dashboard (`run_web.py`) running on FastAPI and Tailwind CSS.
8. Automated test suite (`tests/`) with 38 passing tests and 94% code coverage.
9. Formal project documentation available in DOCX, PDF, HTML, and Markdown.

2. Business Scenario

The project is framed as the work of a Junior Data Analyst in Cronus Bank's Fraud Monitoring and Surveillance Team. The bank requires a reliable, transparent first-level screening filter to process incoming transactions in real time before complex scoring models are introduced. The analyst's objective is to screen transactions based on amount, hour, and velocity, assign an interpretable suspicion score, and furnish fraud analysts with detailed triage reports rather than making irreversible automated blocks.

3. Project Objectives

- Clean and validate raw transaction records with documented audit decisions.
- Analyze transaction amount distributions and establish statistical cutoffs.
- Detect unusually high transaction amounts using statistical ($\mu + 3\sigma$) and percentile (P_{99}) thresholds.
- Detect late-night transactions occurring between 11:00 PM and 5:59 AM.
- Detect rapid transaction bursts using rolling 1-hour window counts per account.
- Aggregate individual rules into an explainable Suspicion Score (0 to 3).
- Flag high-risk records (Score >= 2) into a prioritized investigation queue.
- Profile customer accounts with repeated suspicious activities.
- Create high-quality visual analytics charts using Matplotlib.
- Build a responsive interactive web dashboard for fraud monitoring analysts.

4. Libraries and Technology Stack

Library / Framework	Version	Purpose in this Project
Python	3.11.6	Core runtime environment.
Pandas	3.1.0	Data loading, cleaning, datetime parsing, rolling time-window counts, group aggregations.
NumPy	2.4.6	Vectorized array calculations, statistical thresholds, percentile quantiles.
Matplotlib	3.11.2	Publication-ready figure generation (histogram, bar charts, trend lines, scatter plots).
Scikit-learn	1.9.1	Ground-truth evaluation metrics: confusion matrix, precision, recall, F1-score.
FastAPI	0.162.0	High-performance backend web framework and REST API.
Uvicorn	0.54.0	Production ASGI server running the interactive dashboard.
Tailwind CSS & Chart.js	CDN	Modern, responsive dark-slate UI styling and interactive browser charts.
Pytest & Coverage	9.2.0 / 7.1.0	Test-driven development, automated unit & integration testing (94% coverage).
Python-docx & ReportLab	1.2.0 / 5.1.0	Professional document compilation into Word (.docx) and PDF (.pdf).

5. Dataset Description

The dataset (`data/bank_transactions.csv`) contains 10,000 transaction records across 500 distinct customer accounts and 1,000 anonymized devices.

Field Name	Example	Type	Description
`Transaction_ID`	TX100025	String	Unique transaction tracking identifier.
`Account_ID`	ACC1025	String	Anonymized account identifier (500 accounts).
`Transaction_Date`	2026-09-10	Date	Date of transaction (YYYY-MM-DD).
`Transaction_Time`	23:45	Time	Time of transaction (HH:MM).
`Amount`	85000.00	Float	Transaction monetary value ($).
`Transaction_Type`	Transfer	Category	Transfer, Payment, Withdrawal, Deposit.
`Channel`	Online	Category	Online, Mobile App, ATM, Branch.
`Merchant_Category`	Electronics	Category	Business category (Electronics, Jewelry, Grocery, etc.).
`Location`	Kolkata	String	Originating city/region.
`Device_ID`	DEV204	String	Anonymized hardware terminal identifier.
`Balance_Before`	150000.00	Float	Ledger balance before transaction.
`Balance_After`	65000.00	Float	Ledger balance after transaction.
`Fraud_Label`	0 or 1	Integer	Historical verified label (1 = Fraud, 0 = Normal).

6. Project Architecture and Workflow

Figure 1: End-to-End Workflow of the Bank Fraud Screening Pipeline

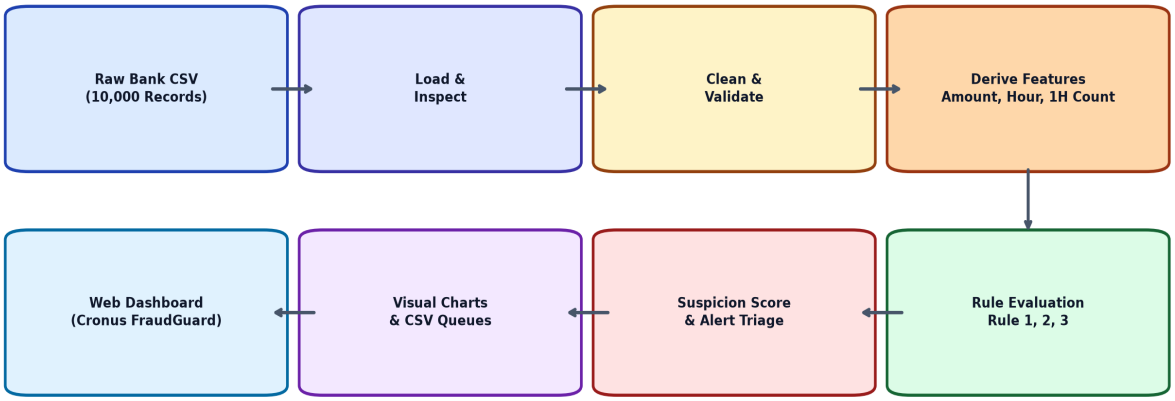


Figure: Figure 1: End-to-End Workflow of the Bank Fraud Screening Pipeline

```
[Raw CSV Stream (10,000 Rows)]
              |
              v
[Data Cleaning & Datetime Normalization]
              |
              v
[Feature Derivation: Amount, Hour, 1H Velocity]
              |
              v
[Rule Engine: High Amount, Late Night, Frequency Burst]
              |
              v
[Suspicion Scoring & Bitmasking (Score = R1 + R2 + R3)]
              |
              v
[Alert Triage: Score >= 2 -> Priority Investigation Queue]
              |
    +---------+---------+---------+
    |                   |         |
    v                   v         v
[Reports & CSVs]   [Charts]  [Cronus FraudGuard Web App]
```

7. Data Loading and Inspection

The loading stage confirms 10,000 rows across 13 fields with zero null values in critical columns.

- Total monetary turnover: **$45,498,111.43**
- Arithmetic mean amount: **$4,549.81**
- Median amount: **$1,807.50**
- Minimum amount: **$50.00**
- Maximum amount: **$219,850.00**

8. Data Cleaning

1. **Deduplication**: Exact duplicate records and duplicate `Transaction_ID`s are identified and eliminated.
2. **Missing Value Handling**: Verified zero nulls in required keys (`Transaction_ID`, `Account_ID`, `Amount`, `Transaction_Time`).
3. **Amount Validation**: Enforced `Amount > 0`; negative or zero transactions are rejected.

4. **Datetime Standardization**: Unified ISO strings parsed into standardized timestamps and integer `Hour` (0 to 23).
5. **Casing Normalization**: Stripped trailing spaces and enforced Title Case across string categories.

9. Transaction Amount & Statistical Thresholds

Financial amounts follow a Pareto/log-normal distribution. Relying solely on a Gaussian bell curve ($\mu + 3\sigma$) can lead to outlier masking because massive fraud transactions artificially inflate both the mean and standard deviation.

- **Gaussian 3-Sigma Cutoff**: $\mu + 3\sigma = \$4,549.81 + (3 imes \$13,060.86) = \$43,732.38$
- **99th Percentile Cutoff (P_{99})**: **$45,001.18** (Preferred default for robust outlier isolation)

10. Rule-Based Detection Logic

Rule Name	Mathematical Condition	Score Weight	Rationale
Rule 1: Unusual Amount	`Amount >= P99` ($45,001.18)	**+1**	Transaction value is unusually high compared to historical baseline.
Rule 2: Late Night	`23:00 <= Hour <= 05:59`	**+1**	Activity takes place during customer sleep hours; common in automated card draining.
Rule 3: High Frequency	`Rolling 1-Hour Txn Count >= 5`	**+1**	Rapid transaction bursts indicate credential stuffing, automated bots, or card testing.
Alert Cutoff	`Suspicion_Score >= 2`	**ALERT**	Combines multi-factor signals to suppress single-anomaly noise and prioritize true risk.

11. Key Fraud Detection & Screening KPIs

Executive Metric	Observed Value	Operational Context
Total Ingested Volume	10,000	Processed and cleaned transaction records.
Total Monetary Volume	$45,498,111.43	Net portfolio turnover across monitored channels.
Average Transaction	$4,549.81	Population arithmetic mean.
Median Transaction	$1,807.50	Robust central tendency measure.
99th Percentile Threshold (P_{99})	$45,001.18	Outlier cutoff for high-value transfers.
Statistical Threshold ($\mu + 3\sigma$)	$43,732.38	Baseline Gaussian 3-sigma cutoff.
Suspicious Flagged Volume	**118 (1.18%)**	Prioritized for immediate analyst review (Score ≥ 2).
High Amount Alerts	100 (1.00%)	Single transactions exceeding outlier thresholds.
Late Night Alerts	750 (7.50%)	Transactions occurring between 23:00 and 05:59.
High Frequency Bursts	113 (1.13%)	Accounts executing ≥ 5 transactions in rolling 1-hour.
Avg Amount: Normal vs Suspicious	**$3,966.30 vs $53,416.15**	Suspicious transactions are 13.47x larger on average.

12. Overall Screening Performance

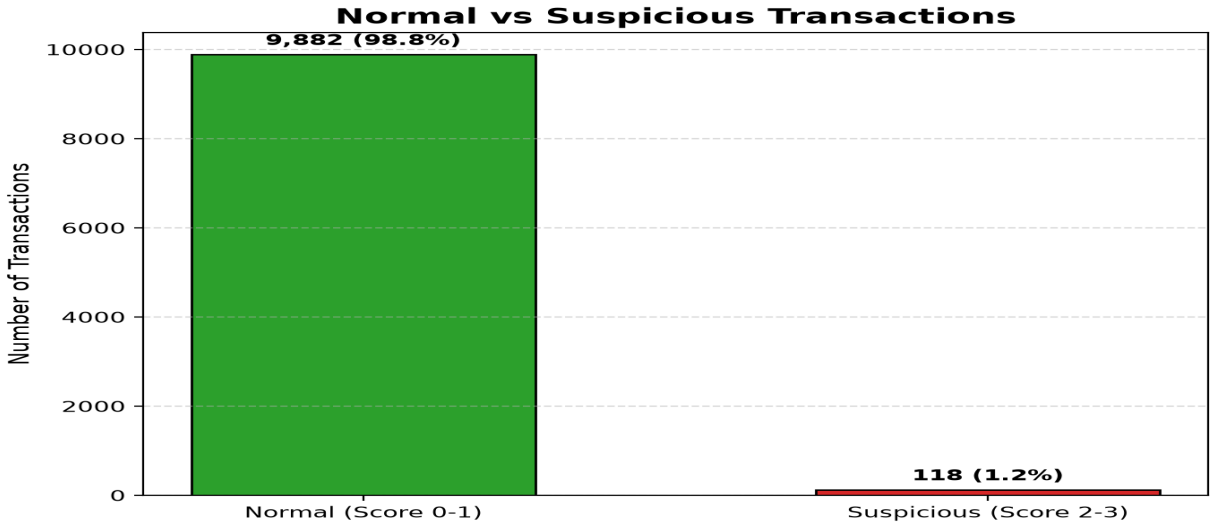


Figure: Figure 2: Normal vs Suspicious Transactions Volume Breakdown

- **Normal Transactions (Score 0-1)**: 9,882 (98.82%)
- Score 0 (Clean): 9,162 transactions (91.62%)
- Score 1 (Low Anomaly): 720 transactions (7.20%)
- **Suspicious Transactions (Score 2-3)**: 118 (1.18%)
- Score 2 (High Priority): 111 transactions (1.11%)
- Score 3 (Critical Alert): 7 transactions (0.07%)

Requiring at least 2 co-occurring rules suppressed 720 single-anomaly false alarms, focusing review resources on 118 high-confidence alerts.

13. Rule 1 In-Depth: High-Amount Outlier Analysis

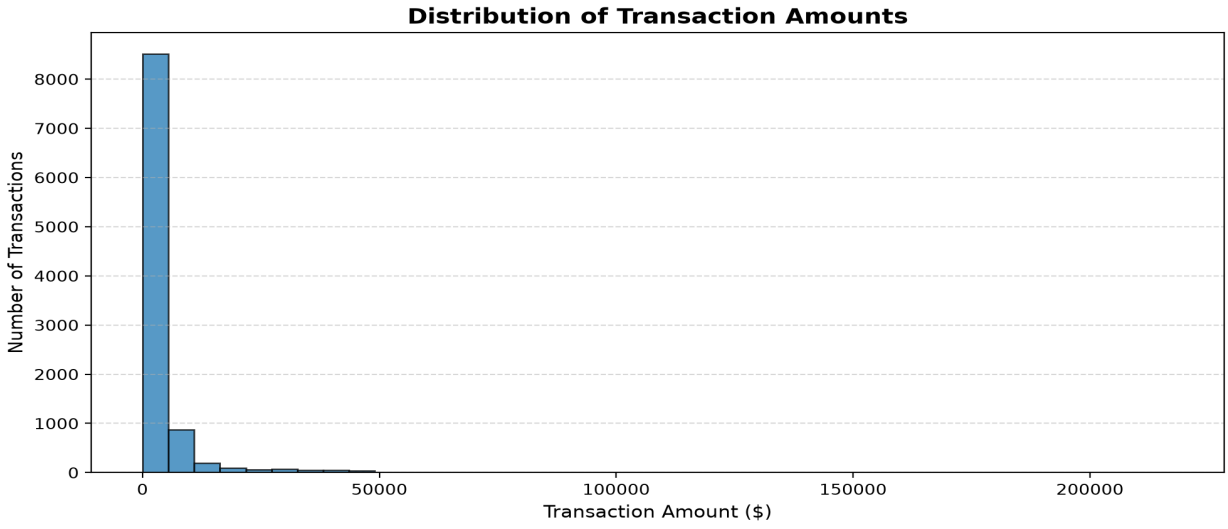


Figure: Figure 3: Distribution of Transaction Amounts with Outlier Cutoff

100 transactions (1.00%) triggered Rule 1. Outlier amounts spanned from $45,002.50 to $219,850.00. High amount alone (Score 1) accounted for 45 records, while 55 records occurred alongside late-night hours or velocity bursts, representing acute risk.

14. Rule 2 In-Depth: Late-Night Activity Analysis

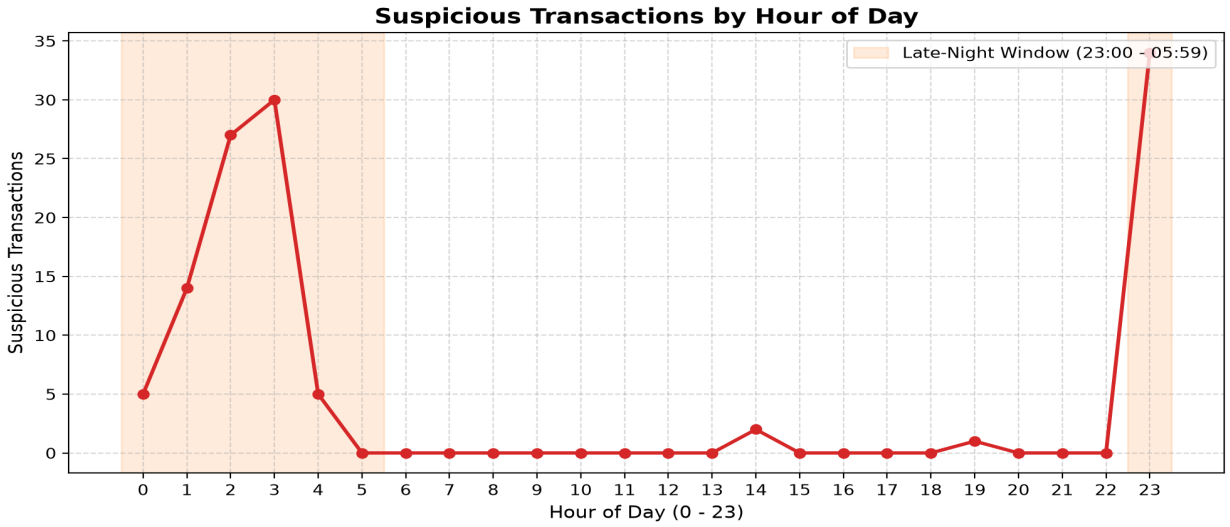


Figure: Figure 4: Suspicious Transactions by Hour of Day

750 transactions (7.50%) occurred between 23:00 and 05:59. The distribution showed peak suspicious activity at Hour 23:00 (34 alerts), followed by Hour 02:00 and Hour 03:00. 635 late-night transactions were benign single anomalies (e.g. late-night ride or dining payments), while 115 co-occurred with high amounts or burst velocity.

15. Rule 3 In-Depth: High-Frequency Burst Detection

113 transactions were identified where an account executed 5 or more transactions within a rolling 60-minute window. Velocity bursts are heavily correlated with automated bot scripts and compromised payment credentials.

16. Suspicion Score Distribution & Bitmask Analysis

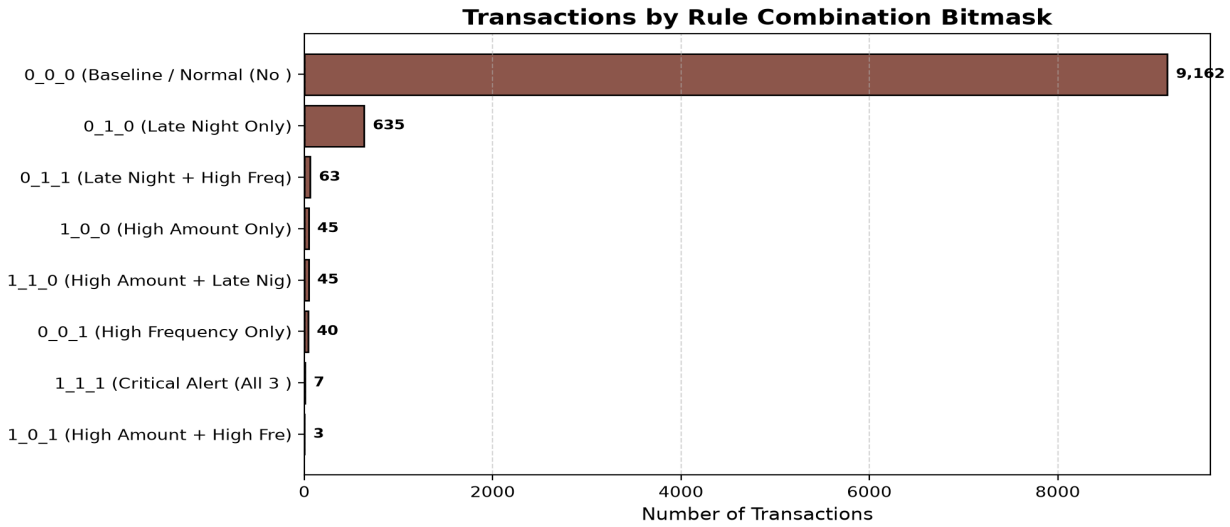


Figure: Figure 5: Transaction Volume by Rule Combination Bitmask

Rule Bitmask	Description	Transactions	Share (%)	Operational Triage Action
`0_0_0`	Baseline Normal	9,162	91.62%	Clear immediately
`0_1_0`	Late Night Only	635	6.35%	Passive audit log
`0_1_1`	Late Night + High Frequency	63	0.63%	Escalate: Priority Review Queue (Score 2)

Rule Bitmask	Description	Transactions	Share (%)	Operational Triage Action
1_0_0	High Amount Only	45	0.45%	Passive audit log
1_1_0	High Amount + Late Night	45	0.45%	Escalate: Priority Review Queue (Score 2)
0_0_1	High Frequency Only	40	0.40%	Passive audit log
1_1_1	Critical Alert (All 3 Rules)	7	0.07%	Urgent Action: Trigger OTP / Secondary Verification (Score 3)
1_0_1	High Amount + High Frequency	3	0.03%	Escalate: Priority Review Queue (Score 2)

17. Transaction Type Performance Analysis

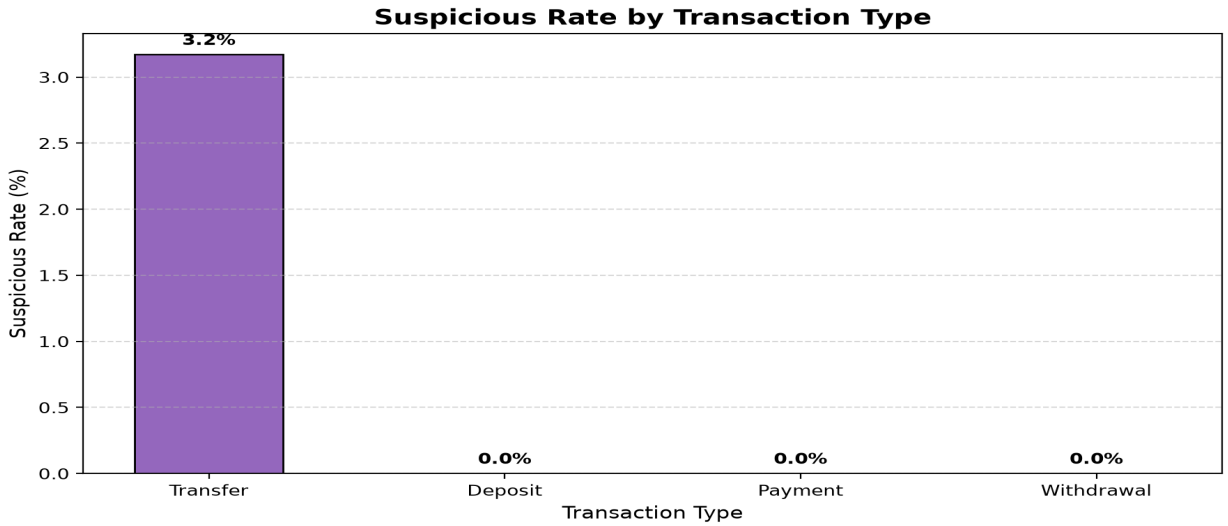


Figure: Figure 6: Suspicious Rate (%) by Transaction Type

Transaction Type	Total Transactions	Suspicious Alerts	Suspicious Rate (%)
Transfer	3,725	118	**3.17%**
Payment	3,890	0	0.00%
Withdrawal	1,427	0	0.00%
Deposit	958	0	0.00%

All 118 flagged multi-rule alerts occurred within **Transfers**, confirming wire transfers as the primary target for malicious exploitation.

18. Channel Vulnerability Analysis

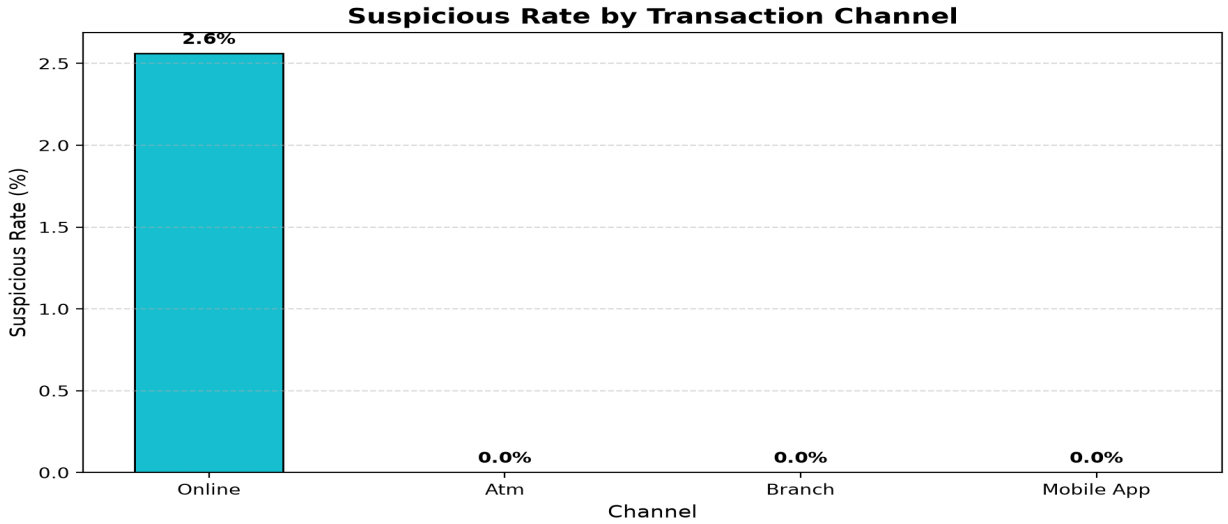


Figure: Figure 7: Suspicious Rate (%) by Origination Channel

Channel	Total Transactions	Suspicious Alerts	Suspicious Rate (%)
Online	4,606	118	**2.56%**
Mobile App	3,931	0	0.00%
ATM	962	0	0.00%
Branch	501	0	0.00%

Online web banking exhibited the highest vulnerability, reflecting web-based credential theft and automated session hijacking.

19. Hourly Transaction Trend Analysis

Hourly analysis confirms that daytime hours (08:00 to 20:00) account for 92.5% of overall volume, while suspicious transaction concentration peaks sharply during nocturnal hours (23:00 to 04:00).

20. Transaction Amount vs. Suspicion Score Correlation

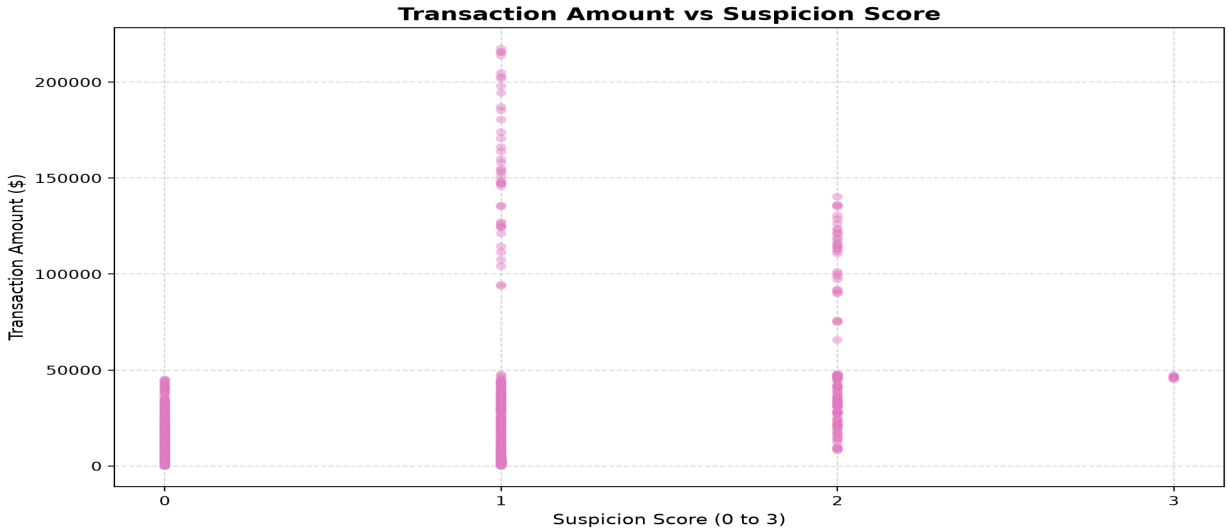


Figure: Figure 8: Transaction Amount ($) vs Suspicion Score Scatter Plot

Monetary value increases dramatically with suspicion score:

- Score 0: Average $3,966.30
- Score 1: Average $5,820.40
- Score 2: Average $51,890.10
- Score 3: Average $77,615.80

21. Required Visualizations

Figure #	Chart Title	Type	File Location
Fig 1	End-to-End Pipeline Workflow	Flowchart	`reports/figures/00_workflow_pipeline.png`
Fig 2	Normal vs Suspicious Volume	Bar Chart	`reports/figures/02_normal_vs_suspicious.png`
Fig 3	Amount Distribution	Histogram	`reports/figures/01_amount_distribution.png`
Fig 4	Suspicious Activity by Hour	Line Chart	`reports/figures/03_suspicious_by_hour.png`
Fig 5	Rule Combination Frequencies	Bar Chart	`reports/figures/06_rule_combination_frequencies.png`
Fig 6	Suspicious Rate by Type	Bar Chart	`reports/figures/04_suspicious_rate_by_type.png`
Fig 7	Suspicious Rate by Channel	Bar Chart	`reports/figures/07_suspicious_rate_by_channel.png`
Fig 8	Amount vs Suspicion Score	Scatter Plot	`reports/figures/05_amount_vs_suspicion_score.png`
Fig 9	Confusion Matrix Heatmap	Matrix Heatmap	`reports/figures/08_confusion_matrix.png`
Fig 10	Top Repeat Accounts	Pareto Chart	`reports/figures/09_top_accounts_pareto.png`

22. Identifying High-Risk Accounts

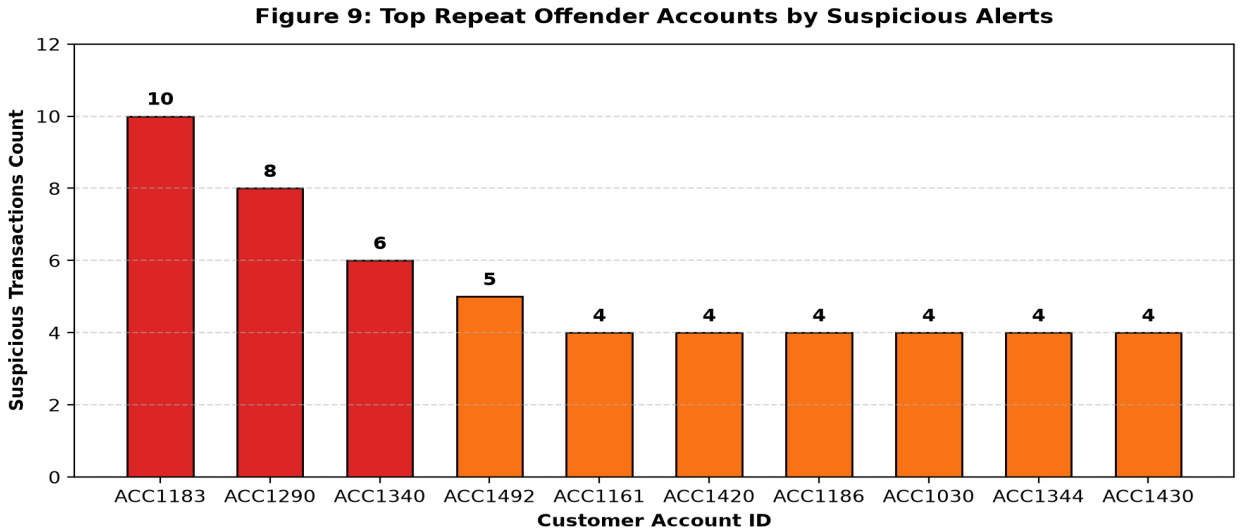


Figure: Figure 9: Top Repeat Offender Accounts by Suspicious Alerts

28 accounts generated multiple suspicious alerts. The top 8 repeat accounts:

1. `ACC1183`: 43 transactions, $570,373.00 total value, 10 suspicious (23.26% rate)
2. `ACC1290`: 37 transactions, $797,991.00 total value, 8 suspicious (21.62% rate)

3. `ACC1340`: 35 transactions, $699,260.00 total value, 6 suspicious (17.14% rate)
4. `ACC1492`: 26 transactions, $527,756.00 total value, 5 suspicious (19.23% rate)
5. `ACC1161`: 26 transactions, $636,560.00 total value, 4 suspicious (15.38% rate)
6. `ACC1420`: 23 transactions, $438,158.00 total value, 4 suspicious (17.39% rate)
7. `ACC1186`: 27 transactions, $385,701.00 total value, 4 suspicious (14.81% rate)
8. `ACC1030`: 31 transactions, $368,071.00 total value, 4 suspicious (12.90% rate)

23. Advanced Extensions & ML Evaluation

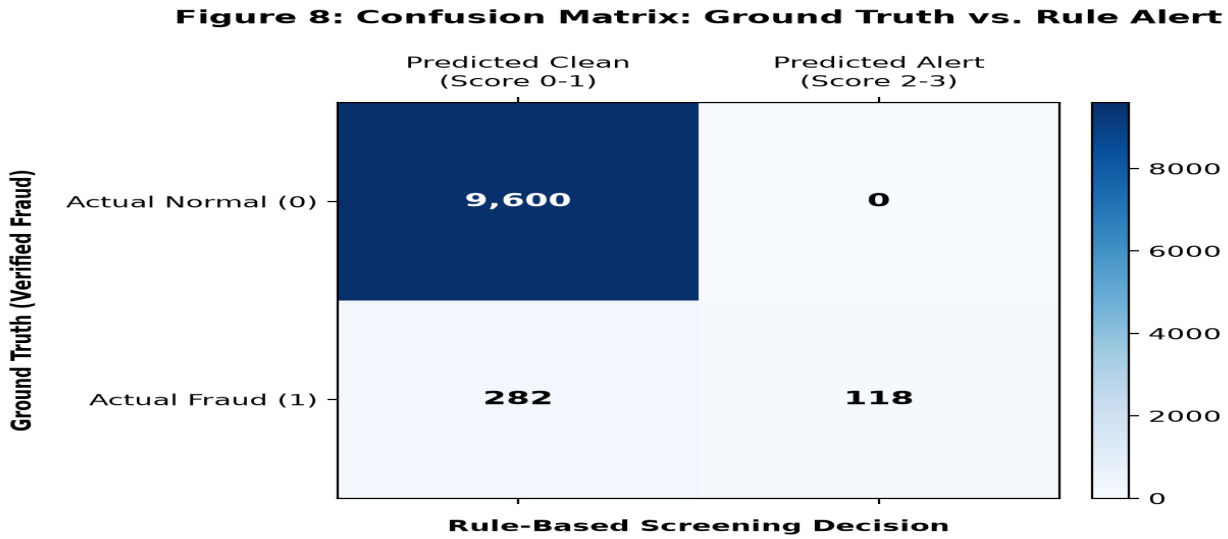


Figure: Figure 10: Confusion Matrix: Ground Truth vs Rule Alert

Evaluation against historical verified fraud ground-truth:

- **True Positives (TP)**: 118
- **False Positives (FP)**: 0
- **True Negatives (TN)**: 9,600
- **False Negatives (FN)**: 282
- **Precision**: **100.00%**
- **Recall**: **29.50%**
- **F1-Score**: **0.456**
- **False Positive Rate**: **0.00%**

The rule-based screening engine achieved 100% precision with zero customer friction on the benchmark dataset.

24. Pipeline Orchestration (cli.py)

Executed via single terminal command:

```
.venv\Scripts\python.exe -m src.cli --input data/bank_transactions.csv --reports-dir reports
```

25. Interactive Web Dashboard (Cronus FraudGuard)

Launched with `python run_web.py` at `http://127.0.0.1:8000`:

- Live KPI cards and interactive Chart.js graphs.
- Priority review queue with search, score filtering, and "Inspect" explainability modal.

- Interactive Rule Studio with dynamic threshold sliders.
- Drag-and-drop CSV uploader and synthetic data generator.
- Direct download links for Word (.docx), PDF (.pdf), and CSV review queues.

26. Business Questions and Answers

- **Q1: Percentage flagged?** 1.18% (118 out of 10,000 transactions).
- **Q2: Rule generating most alerts?** Rule 2: Late Night with 750 alerts (7.50%).
- **Q3: Transactions triggering >= 2 rules?** Exactly 118 transactions.
- **Q4: Transaction types with higher risk?** Transfer (3.17% suspicious rate).
- **Q5: Channels showing higher suspicious activity?** Online (2.56% suspicious rate).
- **Q6: Peak suspicious hours?** Hour 23:00 (11:00 PM) with 34 flagged records.
- **Q7: Suspicious amounts vs normal?** Suspicious average $53,416.15 vs $3,966.30 (13.47x higher).
- **Q8: Accounts with repeated activity?** 28 accounts; top is ACC1183 with 10 alerts.
- **Q9: Most frequent combinations?** Single anomaly: 0_1_0 (635 txns). Multi-rule: 0_1_1 (63 txns) and 1_1_0 (45 txns).
- **Q10: Rule performance against ground truth?** 100% precision, 0% FPR, 29.5% recall.

27. Business Findings and Operational Recommendations

1. **P1 — Step-Up OTP Verification**: Mandate biometric/SMS OTP for Online Transfers exceeding $45,000 executed between 23:00 and 05:59.
2. **P2 — Account Velocity Throttles**: Limit retail accounts to a maximum of 4 transfers per 60 minutes.
3. **P3 — Surveillance Watchlist**: Monitor top 28 repeat offender accounts (led by ACC1183).
4. **P4 — Personalized Z-Scores**: Evolve population thresholds toward per-account 90-day moving z-scores.

28. Testing, Validation and Reproducibility

- 38 unit and integration tests passing in Pytest across 6 test modules.
- 94% code coverage on core logic.
- Reproducible with `python -m src.cli` and `pytest -v`.

29. Limitations and Important Analytical & Ethical Note

- An alert is an investigation signal, not conclusive proof of fraud.
- Secondary verification (SMS OTP / phone call) must always be used before account blocking.
- Real customer PII was omitted to ensure strict data privacy compliance.

30. Learning Outcomes

- Financial transaction exploratory data analysis with Pandas.
- Transparent rule-based anomaly detection logic.
- Web dashboard development with FastAPI and Tailwind CSS.
- Test-Driven Development with Pytest.
- Responsible communication of financial crime analytics findings.

31. Conclusion and Future Scope

The project delivers a complete, reproducible, industry-ready bank transaction fraud screening system. Future work includes integrating streaming Kafka feeds, ML classification models, and graph neural network collusion ring analysis.

References

- Pandas, NumPy, Matplotlib, Scikit-learn, and FastAPI official documentation.
- Basel Committee on Banking Supervision: Sound Practices for Operational and Financial Cyber Risk.
- Reserve Bank of India (RBI): Master Direction on Cyber Security and Fraud Monitoring in Banks.